

Proximal Policy Optimization (PPO): Formal Definitions and Equations

1 Core Reinforcement Learning Quantities

1.1 Trajectory

The trajectory τ is the sequence of states, actions, and rewards generated by interacting with the environment under a policy.

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_T, a_T, r_T)$$

1.2 Policy (Actor)

The policy is a parameterized probability distribution over actions given a state.

$$\pi_\theta(a_t \mid s_t)$$

1.3 Total Discounted Return

The cumulative discounted future reward from time t onward.

$$G_t = \sum_{l=0}^{T-t} \gamma^l r_{t+l}$$

1.4 State-Value Function (Critic)

Expected return when starting from state s_t and following policy π .

$$V^\pi(s_t) = \mathbb{E}_\pi[G_t \mid s_t]$$

Function approximation used in PPO:

$$V_\phi(s_t)$$

1.5 Action-Value Function

Expected return after taking action a_t in state s_t .

$$Q^\pi(s_t, a_t) = \mathbb{E}_\pi[G_t \mid s_t, a_t]$$

1.5.1 One-Step Bootstrapped Approximation

$$\hat{Q}(s_t, a_t) = r_t + \gamma V_\phi(s_{t+1})$$

1.6 Advantage Function

Measures how much better an action is compared to the expected value of the state.

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$$

1.7 Temporal Difference (TD) Error

One-step prediction error between successive value estimates.

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

1.8 Generalized Advantage Estimation (GAE)

Recursive form:

$$A_t^{GAE} = \delta_t + \gamma \lambda A_{t+1}^{GAE}$$

Expanded form:

$$A_t^{GAE} = \sum_{l=0}^{T-t} (\gamma \lambda)^l \delta_{t+l}$$

1.9 Target Value for Critic

Learning target used to update the value function.

$$\hat{V}_t = A_t^{GAE} + V_\phi(s_t)$$

2 PPO Optimization Components

2.1 Probability Ratio

Relative change between updated and behavior policy.

$$r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$$

2.2 Clipped Surrogate Objective

$$L^{\text{clip}}(\theta) = \mathbb{E}_t [\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$$

2.3 Value Function Loss

$$L^{VF}(\phi) = \mathbb{E}_t \left[\left(V_\phi(s_t) - \hat{V}_t \right)^2 \right]$$

2.4 Entropy Bonus

Encourages exploration via policy stochasticity.

$$L^{entropy}(\theta) = \mathbb{E}_t [H(\pi_\theta(\cdot \mid s_t))]$$

2.5 Full PPO Objective

Maximization form:

$$L^{total} = L^{clip} - c_1 L^{VF} + c_2 L^{entropy}$$

Minimization form:

$$\mathcal{L}_{PPO} = -L^{clip} + c_1 L^{VF} - c_2 L^{entropy}$$

3 Actor-Critic Architecture

Actor (policy):

$$\pi_\theta(a \mid s)$$

Critic (value function):

$$V_\phi(s)$$